

Homework — 2.1.4 Thinking Logically — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. (a) [1] `height >= 1.4` (accept `height >= 140` if cm stated; accept an `IF` wrapper). Must use `>=` — "at least" includes exactly 1.4 m, so `>` earns nothing.

(b) [2] 1 mark TRUE: the rider is allowed on / admitted. 1 mark FALSE: the rider is refused / turned away.

2. [2] 1 mark condition: `amount <= balance AND amount <= dailyLimit` (accept equivalent variable names). 1 mark FALSE outcome: the withdrawal is **refused/declined** and no cash is dispensed. The condition must use the **AND** operator.

3. (a) [2] 1 mark: `level < 20` (THEN pump ON). 1 mark: `level > 80` (THEN pump OFF). (Strict `<` / `>` match "falls below" / "rises above"; `<=` / `>=` earn nothing here.)

(b) [1] The pump is left **unchanged** / no action is taken — neither condition is true at 50%, so neither branch is followed.

4. [2] 1 mark: under `spent > 100`, a customer who has spent exactly £100 does **not** qualify (100 is not greater than 100). 1 mark: rewrite as `spent >= 100`.

5. [4] Up to 4 marks: - First decision `score >= 70` (1); TRUE branch → Distinction (1). - Second decision, tested only when the first is FALSE: `score >= 40` (1); TRUE branch → Pass, ELSE branch → Fail (both needed) (1). - The conditions must be **ordered/nested** so that 70+ is tested first (credited within the marks above). Conditions must be exact Booleans applied to this scenario; vague wording such as "if the score is high" earns no condition mark.

Part B (6 marks)

6. [2] 1 mark: the tax step uses the **gross pay** value produced by the previous step as its input. 1 mark: because of this data dependency it cannot run until gross pay has been calculated — the order of the two steps is fixed.

7. [2] 1 mark: a **primary key** is an attribute (or set of attributes) that **uniquely identifies** each record in a table. 1 mark: a **foreign key** is an attribute in one table that refers to the primary key of another table, linking the two tables.

8. [2] `00101101`. 1 mark for the correct binary value (101101), 1 mark for the correct 8-bit form with leading zeros. ($45 = 32 + 8 + 4 + 1$.)

Total: 14 + 6 = 20 marks